

Many human traits follow Mendelian patterns of inheritance

The study of human genetics is fueled by our desire to understand our own inheritance and to develop treatments for genetically based diseases. While peas are convenient subjects for genetic research, humans are not. The human generation span is long (about 20 years), human parents produce many fewer offspring than peas, and breeding experiments would be unethical!

Pedigree Analysis

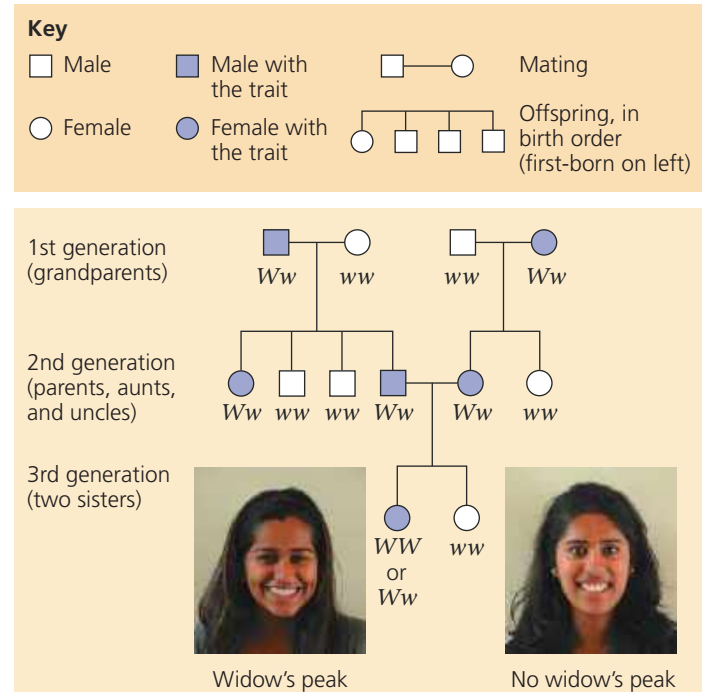
In place of breeding experiments, geneticists analyze the results of human matings that have already occurred. They collect information about a family's history for a particular trait and assemble this information into a family tree describing the trait across the generations—a family **pedigree**.

Figure 14.15a shows a three-generation pedigree that traces the occurrence of a pointed contour of the hairline on the forehead. We simplify the genetics of this trait, called a widow's peak, by assuming it is due to a dominant allele, W . (In reality, other genes are probably also involved.) In our model, because the widow's peak allele is dominant, all individuals who lack a widow's peak must be homozygous recessive (ww). The two grandparents with widow's peaks must have the Ww genotype since some of their offspring are homozygous recessive. The offspring in the second generation who *do* have widow's peaks must also be heterozygous because they are the products of $Ww \times ww$ matings. The third generation in this pedigree consists of two sisters. The one who has a widow's peak could be either homozygous (WW) or heterozygous (Ww), given what we know about the genotypes of her parents (both Ww).

Figure 14.15b is a pedigree of the same family, but this time we focus on a recessive trait, the inability of individuals to taste a chemical called PTC (phenylthiocarbamide). Compounds similar to PTC are found in broccoli, brussels sprouts, and related vegetables and account for the bitter taste some people report when eating these foods. We'll use t for the recessive allele and T for the dominant allele, which results in the ability to taste PTC. As you work your way through the pedigree, notice once again that you can apply what you have learned about Mendelian inheritance to understand the genotypes shown for the family members.

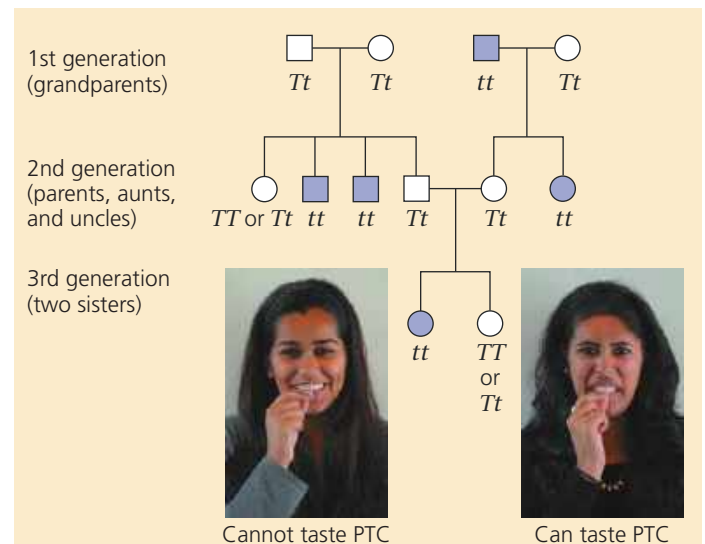
An important application of a pedigree is to help us calculate the probability that a future child will have a particular genotype and phenotype. Suppose that the couple represented in the second generation of Figure 14.15 decides to have one more child. What is the probability that the child will have a widow's peak? This is equivalent to a Mendelian F_1

▼ **Figure 14.15 Pedigree analysis.** Each pedigree traces a trait through three generations of the same family; the two traits have different inheritance patterns. For the sake of understanding genetic principles, each trait is shown as being determined by two alleles of a single gene—a simplification, since other genes likely also affect these two characters.



(a) Is a widow's peak a dominant or recessive trait?

Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), *all* of their offspring would also have the recessive phenotype.



(b) Is the inability to taste a chemical called PTC a dominant or recessive trait?

Tips for pedigree analysis: Notice that the first-born daughter in the third generation has the trait (is unable to taste PTC), although both parents lack that trait (they *can* taste PTC). Such a pattern is explained if the non-taster phenotype is due to a recessive allele. If it were due to a dominant allele, then at least one parent would also have had the trait.